

SUMMARY

This video explains the the principles of Compound Finite Automata with emphasis on minimal DFA state calculation under multiple divisibility constraints. The core distinction lies between:

Independent symbol constraints (require cross multiplication of states)

Shared property constraints (length or single-symbol counts, require LCM/GCD logic)

Key conclusions:

AND operations use LCM.

OR operations use GCD only when one divisor divides the other.

Complementation does not change the number of states.

1. MULTI-SYMBOL CONSTRAINTS AND CROSS MULTIPLICATION

When divisibility constraints apply to different symbols in the alphabet, state counts multiply.

1.1 Case Study: Divisibility of a, b, and c

Given:

Number of a's divisible by 2 $\rightarrow$ 2 states

Number of b's divisible by 3 $\rightarrow$ 3 states

Number of c's not divisible by 5 $\rightarrow$ 5 states

Even though the condition for c is "not divisible," the modulo-5 structure still requires 5 states. Complementation only changes final states, not total states.

Total States = $2 \times 3 \times 5 = 30$

2. SHARED PROPERTY CONSTRAINTS (LENGTH OR SINGLE SYMBOL)

When constraints apply to the same property (e.g., total length or number of a's), cross multiplication does not apply. Instead, LCM and GCD rules determine the state count.

TYPE A: ONE DIVISOR DIVIDES THE OTHER

Condition: m divides n

Example: 2 and 4

Operation	Minimal States
AND	$\text{LCM}(m, n)$
OR	$\text{GCD}(m, n)$

Example:

Divisible by 2 AND 4

$$\text{LCM}(2, 4) = 4$$

Divisible by 2 OR 4

$$\text{GCD}(2, 4) = 2$$

TYPE B: COPRIME DIVISORS ($\text{GCD} = 1$)

Example: 2 and 3, 3 and 7

Operation	Minimal States
AND	$\text{LCM}(m, n) = m \times n$
OR	$\text{LCM}(m, n) = m \times n$

Example:

Divisible by 2 AND 3

$$\text{LCM}(2, 3) = 6$$

Divisible by 4 OR 9

$$\text{LCM}(4, 9) = 36$$

TYPE C: SHARED FACTORS BUT NOT MULTIPLES ($\text{GCD} > 1$)

Example: 4 and 6, 6 and 8

Operation **Minimal States**

AND $\text{LCM}(m, n)$

OR $\text{LCM}(m, n)$

Example:

Divisible by 4 AND 6

$\text{LCM}(4,6) = 12$

Divisible by 4 OR 6

$\text{LCM}(4,6) = 12$

3. AD RULE SUMMARY TABLE

Relationship Between m and n	Operation	Minimal DFA States
m divides n	AND	$\text{LCM}(m, n)$
m divides n	OR	$\text{GCD}(m, n)$
$\text{GCD}(m, n) = 1$	AND	$\text{LCM}(m, n)$
$\text{GCD}(m, n) = 1$	OR	$\text{LCM}(m, n)$
$\text{GCD}(m, n) > 1$ (not multiple)	AND	$\text{LCM}(m, n)$
$\text{GCD}(m, n) > 1$ (not multiple)	OR	$\text{LCM}(m, n)$

Key Insight:

The only scenario where GCD applies is OR under Type A (when one divisor divides the other). In all other shared-property cases, LCM determines the state count.

4. COMPLEMENTATION AND “NOT” CONDITIONS

Complementation does not increase state count.

4.1 Structural Principle

A DFA recognizing:

$n \bmod k = 0$

has the same transition structure as:

$$n \bmod k \neq 0$$

Only the set of final states differs.

4.2 Example (GATE 2020)

Condition:

Number of a's divisible by 2

Number of a's not divisible by 3

Interpretation:

$$n_a \bmod 2 = 0$$

$$n_a \bmod 3 \neq 0$$

Operation:

AND

State Count:

$$\text{LCM}(2,3) = 6$$

Complementation does not change total states.

5. PRACTICAL EXAM STRATEGY

Efficiency Gains:

Using LCM/GCD rules avoids manual DFA construction.

State counts can be determined in seconds.

Time saved can be redirected to lengthy analytical problems.

Consistency:

LCM applies to almost all shared-property constraints.

GCD applies only in OR when one divisor divides the other.

Cross multiplication applies only for independent symbol constraints.

FINAL CONCLUSION ON MINIMAL DFA STATE CALCULATION

Case	Rule
Distinct Symbols (n_a and n_b)	Multiply states ($m \times n$)
Shared Property (Length or single symbol)	Use LCM
OR with divisibility where one divides the other	Use GCD
Complement or "Not" condition	State count unchanged

Minimal DFA state determination becomes systematic once the structural relationship between constraints is identified.